

Probing the two-quasiparticle $K^\pi = 8^+$ isomeric structure and enhanced stability in the proton drip-line nuclei

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Probing the two-quasiparticle $K^\pi = 8^+$ isomeric structure and enhanced stability in the proton drip-line nuclei

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Stimulated by recent experimental discoveries [Phys. Lett. B **847**, 138310 (2023) and Phys. Rev. Lett. **132**, 072502 (2024)], two-quasiparticle $K^\pi = 8^+$ isomeric structure (related to the neutron $h_{9/2}$ and $f_{7/2}$ orbitals) in $^{160}_{76}\text{Os}_{84}$ that lies at the two-proton drip line has been studied by means of the configuration-constrained potential-energy-surface calculations. Calculated results indicate that, for such an isomer, the excitation energy can be well reproduced and its oblate shape can be enhanced by the polarization effects of the two high- K orbits. Comparing with experimental data, two sets of the widely used Woods-Saxon parameters, especially, the spin-orbit coupling one, are evaluated and argued. It is found that, considering the uncertainty of the spin-orbit coupling strength, the energy crossing or inversion of the $h_{9/2}$ and $f_{7/2}$ neutrons can occur, which may lead to three kinds of different evolution-trends of two-quasiparticle excitation energies with the changing quadrupole deformation β_2 . With decreasing spin-orbit coupling interaction, the structure of the $K^\pi = 8^+$ isomeric state will evolve from $\nu h_{9/2} f_{7/2}$ ($\nu 9/2^- [505] \otimes 7/2^- [503]$) to the mixing of $\nu h_{9/2} f_{7/2}$ and $\nu h_{9/2}^2$ ($\nu 9/2^- [505] \otimes 7/2^- [514]$) to $\nu h_{9/2}^2$, indicating that its structural probes is still of interest and an arbitrary assignment may be risky. The related theoretical calculations and experimental evidences e.g., the transition properties, are desirable. In addition, similar to that in superheavy nuclei, it is suggested that the stability inversion between high- K isomeric states and ground states might occur in this proton drip-line mass region, e.g., in the hitherto unknown nucleus $^{162}_{78}\text{Pt}_{84}$.

- ^{160}Os : Two independent experiments [[Phys. Lett. B 847, 138310 \(2023\)](#)] and [[Phys. Rev. Lett. 132, 072502 \(2024\)](#)] used the same fusion-evaporation reaction but different beam energies (310 and 335 MeV, respectively)
- Both reported highly consistent ground-state properties
- The discrepancy on the $K^\pi = 8^+$ Isomer
 - The PLB (2023) experiment clearly identified a two-quasiparticle $K^\pi = 8^+$ isomeric state.
 - The PRL (2024) experiment found absolutely no evidence for this isomer.
- In N=84 isotones, the 8^+ state is systematically driven by the $\nu f_{7/2}h_{9/2}$ configuration. However, the relative positions of the $\nu 2f_{7/2}$ and $\nu 1h_{9/2}$ orbitals are highly sensitive to Spin-Orbit Coupling (λ) uncertainty.

————→ uncertainty [29]. One of our aims in this project is to explore the effects of spin-orbit coupling on the intrinsic configuration of the two-quasiparticle $K^\pi = 8^+$ isomeric states, especially, in $^{160}_{76}\text{Os}_{84}$ and $^{162}_{78}\text{Pt}_{84}$.

Stability Inversion: In superheavy nuclei : High- K isomers can outlive their ground states , providing a pathway for synthesizing superheavy nuclei.

In the drip-line region, it is of interest to probe whether the broken-pair, high- K isomers may confer extra stability.



ity. In particular, to look for the possible candidate for which the stability inversion occurs between the high- K isomeric state and the ground state in this nuclear region will be our another research purpose in this project.

evaluation of their α -decay half lives. We are also interested in testing the applications of phenomenological Woods-Saxon mean-field model and model parameters (e.g., the uncertainty of the spin-orbit coupling strengths and its effect on nuclear configurations).

- Using macroscopic-microscopic method to calculate energy in different deformed parameters to get the constrained PES:

$$E(Z, N, \beta) = E_{\text{macro}}(Z, N, \beta) + E_{\text{micro}}(Z, N, \beta) = E_{\text{LDM}} + E_{\text{LN}} - \tilde{E}_{\text{Stut}}$$

The microscopic part uses the Strutinsky shell correction method.

- Liquid drop Model:

$$E_{\text{LDM}} = a_{\text{vol}}A + a_{\text{surf}}A^{2/3} + a_{\text{coul}}Z^2A^{-1/3} + a_{\text{sym}}\frac{(N - Z)^2}{A}$$

- Deformed Woods-Saxon potential and Lipkin-Nogami method:

$$E_{\text{LN}} = \sum_{j=1}^S e_{k_j} + 2 \sum_{k \neq k_j} v_k^2 e_k - \frac{\Delta^2}{G} - G \sum_{k \neq k_j} v_k^4 + G \frac{N - S}{2} - 4\lambda_2 \sum_{k \neq k_j} (u_k v_k)^2$$

where S is the proton or neutron seniority for a given configuration.

- Calculations were carried out in the (β_2, γ) deformation space, with hexadecapole deformation β_4 included at each point.

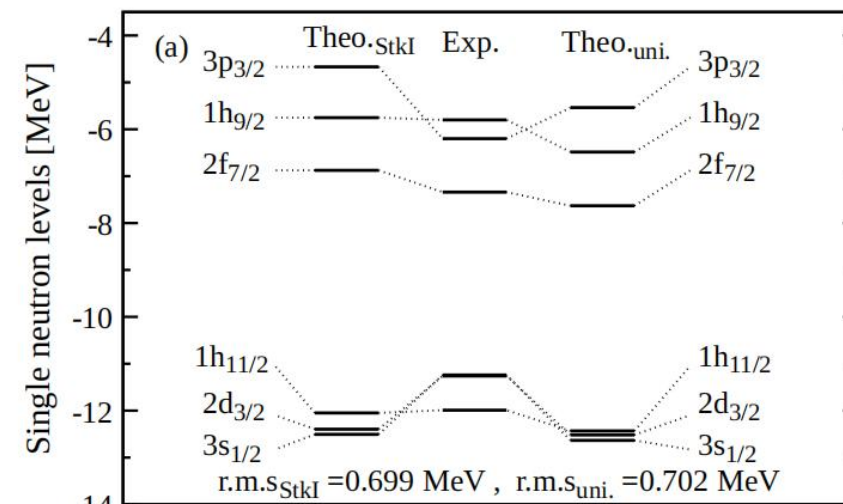
Evaluation and selection of model parameters

- Find the optimal WS parameters for ^{160}Os : Measuring the "distance" (Euclidean, Manhattan, Chebyshev) between ^{160}Os and 8 doubly-magic nuclei.

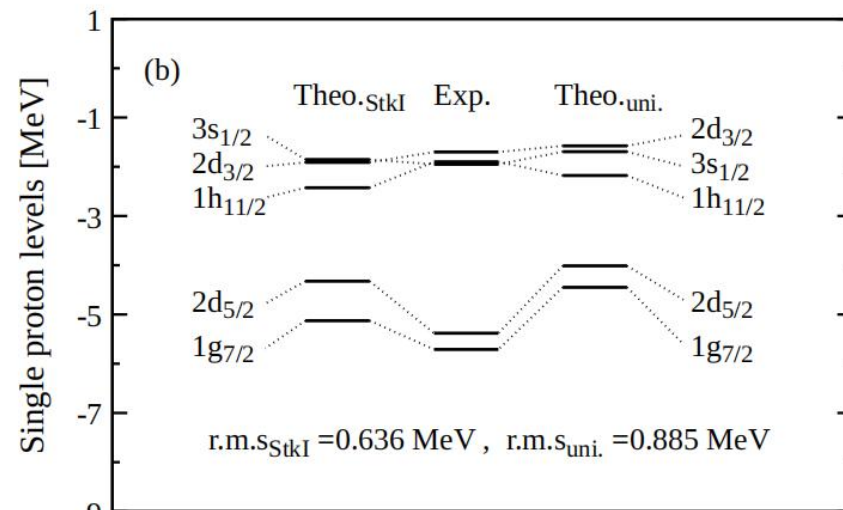
the Minkowski distance $d(P, P') = \sqrt[k]{\sum_{i=1}^n |x_i - x'_i|^k}$.

Nuclei	d_E	d_M	d_C
$^{16}_8\text{O}_8$	102.0	144	76
$^{40}_{20}\text{Ca}_{20}$	85.0	120	64
$^{48}_{20}\text{Ca}_{28}$	79.2	112	56
$^{56}_{28}\text{Ni}_{28}$	73.8	104	56
$^{90}_{40}\text{Zr}_{50}$	49.5	70	36
$^{132}_{50}\text{Sn}_{82}$	26.1	28	26
$^{146}_{64}\text{Gd}_{82}$ ✓	12.2	14	12
$^{208}_{82}\text{Pb}_{126}$	42.4	48	42

- Test on ^{146}Gd : Universal vs. Stockholm-I (StkI).
StkI exhibits smaller r.m.s. deviations and better reproduces the crucial high-j $\nu 1h_{9/2}$ orbital.

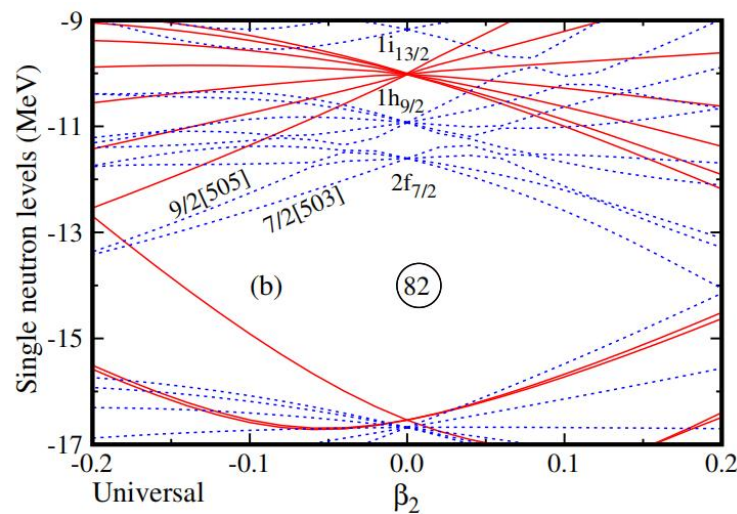
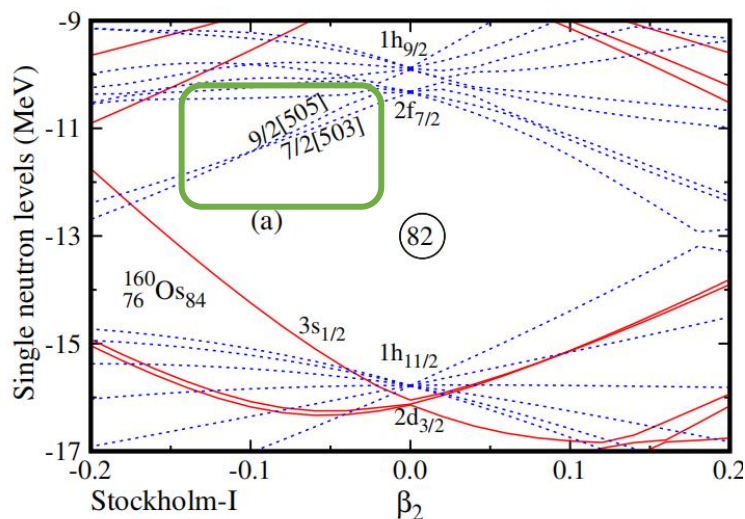


$^{146}_{64}\text{Gd}_{82}$ Spherical Woods-Saxon Hamiltonian



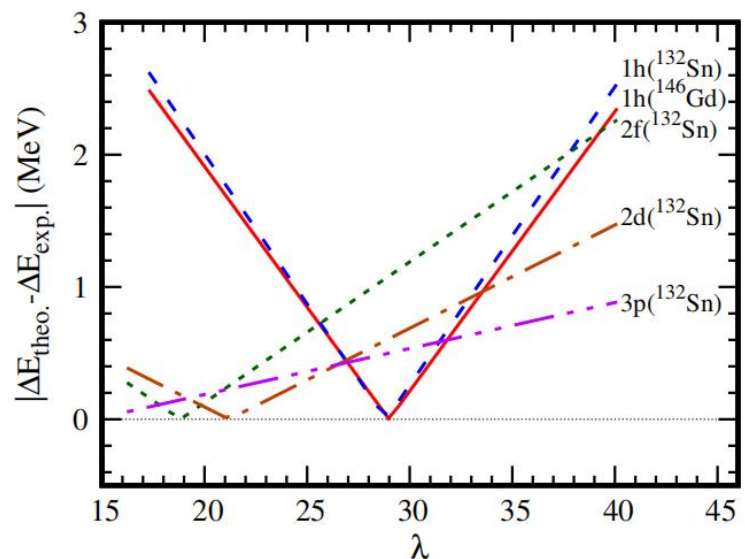
$^{146}_{64}\text{Gd}_{82}$ Spherical Woods-Saxon Hamiltonian

• The Crucial Orbitals in ^{160}Os



along β_2 . Taking the report in Ref. [21] into account, we prefer to **select the StkI parameter set** for the following calculations in this work since it was pointed out that, relative to the $\nu f_{7/2}$ level, the energy of $\nu h_{9/2}$ level has a decreasing trend (namely, the energy difference between $\nu h_{9/2}$ and $\nu f_{7/2}$ orbitals decreases) in the $N = 84$ isotones from $Z = 64$ to 76, even smaller than that of $\nu f_{7/2}$, approaching the present case.

• Impact of Spin-Orbit Coupling (λ)



- $\nu 2f_{7/2}$ ($j=l+1/2$) and $\nu 1h_{9/2}$ ($j=l-1/2$) have opposite responses to the Spin-Orbit Coupling strength (λ). Their relative positions are extremely sensitive to λ .
- Extracting the "Optimal λ " from spin doublets (e.g., the splitting ΔE of 1h, 2f) in ^{146}Gd and ^{132}Sn .

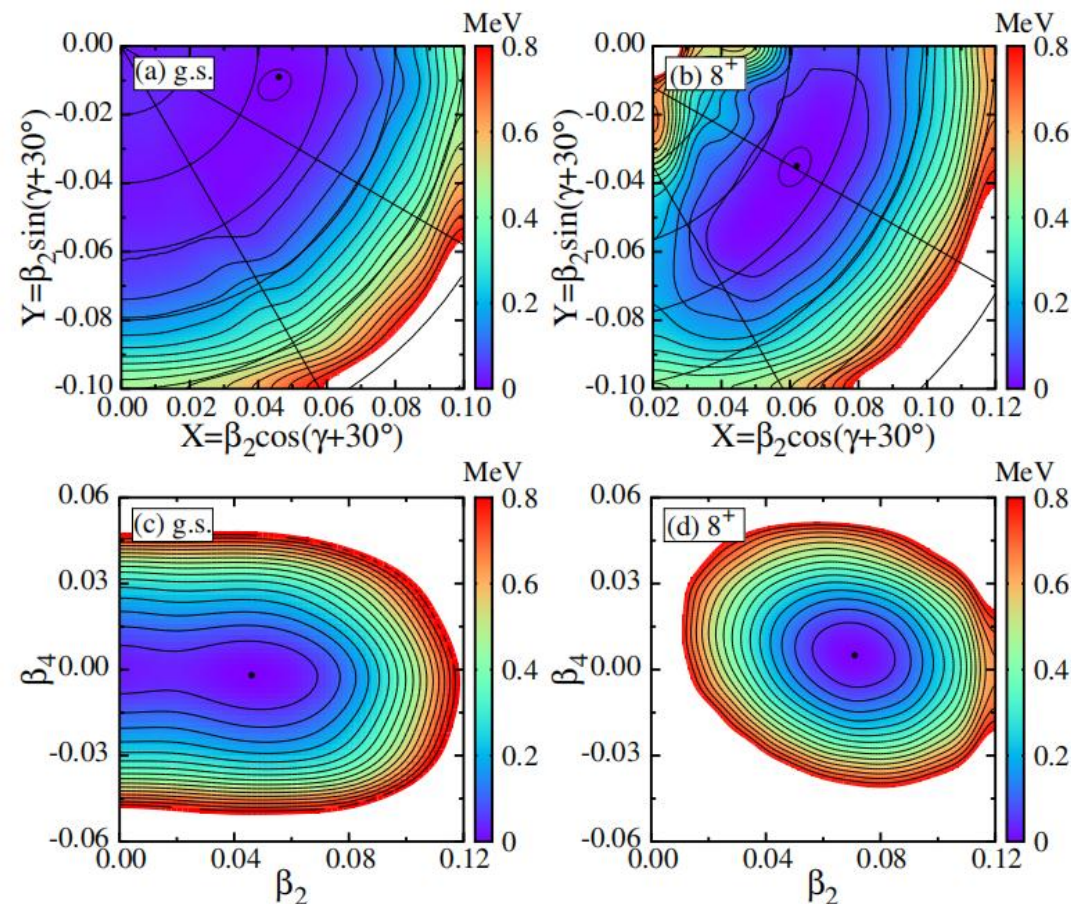
29, 21, 19 and 14, respectively. Though the reliability of such a single parameter adjustment is somewhat restricted, it still bears some important information on the large uncertainty of spin-orbit coupling parameter λ . Recently, in Ref. [48, 55, 56] employing phenomenological Woods-Saxon model, a new spin-orbit strength $\lambda = 26.210(0.513)$ is suggested after removing parametric correlations.

Isomeric structure and its enhanced stability

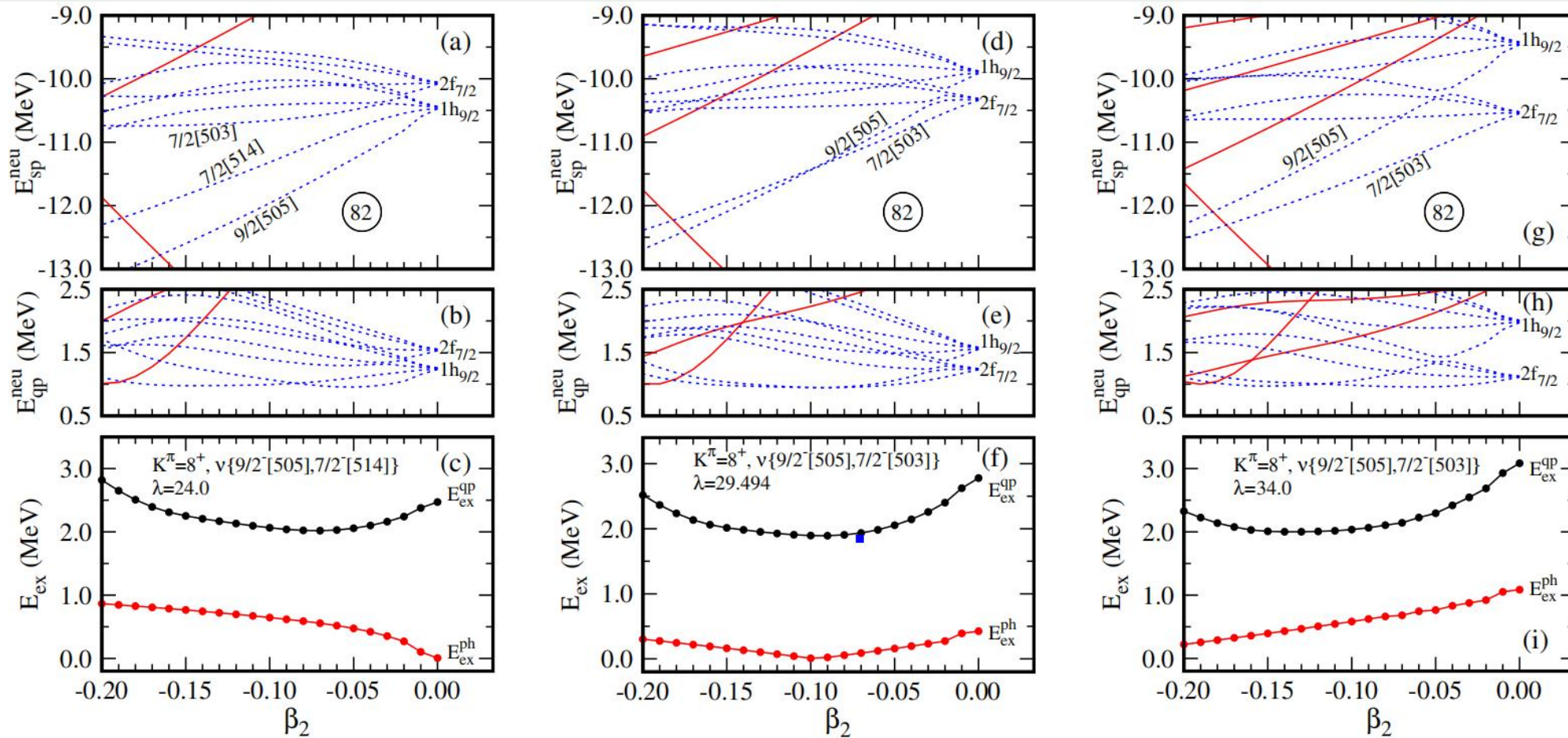
Figure 4 plots the ground-state PESs and the configuration-constrained ones for two-neutron $K^\pi = 8^+$, $7/2^- [503] \otimes 9/2^- [505]$ isomeric state for $^{160}_{76}\text{Os}_{84}$, indicating the equilibrium deformations and the softnesses along different deformation degrees of freedom. Let us

Ground state: Weakly oblate and β_2 -soft

$K^\pi = 8^+$ Isomer: A deeply stabilized oblate shape.



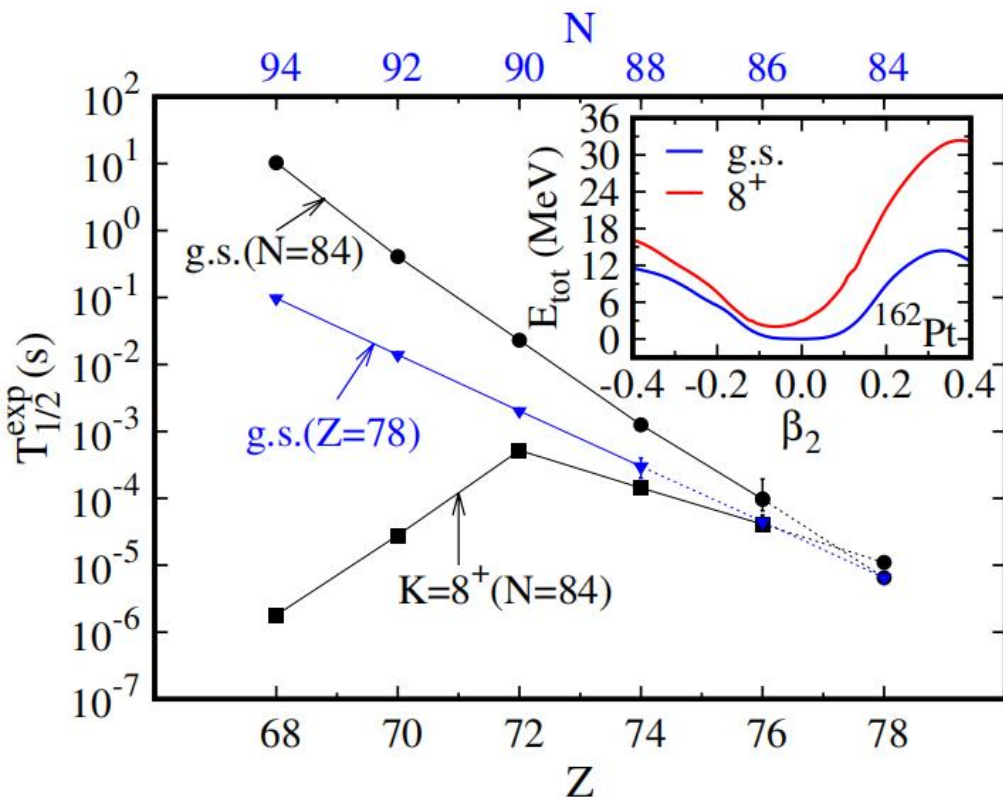
Nuclei	g.s. (this work)			$K^\pi = 8^+$ (this work)				g.s. (Ref. [49])		Exp.(Ref. [24])		
	β_2	γ	β_4	β_2	γ	β_4	E^{ex}	β_2	β_4	E^{ex}	$T_{1/2}(8^+)$	$T_{1/2}(\text{g.s.})$
$^{160}_{76}\text{Os}_{84}$	0.047	41°	-0.003	0.071	60°	0.005	1.885	0.064	-0.010	1.844(0.018)	$41^{+15}_{-9}\mu\text{s}$	$97^{+97}_{-32}\mu\text{s}$
$^{162}_{78}\text{Pt}_{84}$	0.001	56°	0.000	0.062	60°	0.005	2.029	-	-	-	-	-



- (a, d, g) Single-particle levels: Shows the crossing and inversion of crucial orbitals ($\nu 2f_{7/2}$ and $\nu 1h_{9/2}$) under different λ .
- (b, e, h) Quasi-particle levels: Based on single-particle levels, pairing effects are included.
- (c, f, i) Excitation energies: The 1p-1h energies (red lines) are too low. Adding pairing effects (black lines) perfectly reproduces the experimental 1.844 MeV excitation energy (Blue square in Fig. 5f).

λ	$e(\text{MeV})$	Label	First three w.f. components
24.0	-11.26	$ 505\frac{9}{2}\rangle$	$87.8\% 505\frac{9}{2}\rangle + 11.9\% 514\frac{9}{2}\rangle + 0.2\% 1105\frac{9}{2}\rangle$
		$ 5h_{9/2}\frac{9}{2}\rangle$	$99.4\% 5h_{9/2}\frac{9}{2}\rangle + 0.2\% 11h_{9/2}\frac{9}{2}\rangle + 0.2\% 5h_{11/2}\frac{9}{2}\rangle$
	-10.79	$ 514\frac{7}{2}\rangle$	$50.6\% 514\frac{7}{2}\rangle + 44.4\% 503\frac{7}{2}\rangle + 2.1\% 303\frac{7}{2}\rangle$
		$ 5h_{9/2}\frac{7}{2}\rangle$	$56.9\% 5h_{9/2}\frac{7}{2}\rangle + 38.1\% 5f_{7/2}\frac{7}{2}\rangle + 2.7\% 3f_{7/2}\frac{7}{2}\rangle$
	-10.49	$ 503\frac{7}{2}\rangle$	$35.3\% 503\frac{7}{2}\rangle + 31.2\% 523\frac{7}{2}\rangle + 27.8\% 514\frac{7}{2}\rangle$
		$ 5f_{7/2}\frac{7}{2}\rangle$	$51.0\% 5f_{7/2}\frac{7}{2}\rangle + 42.5\% 5h_{9/2}\frac{7}{2}\rangle + 3.0\% 3f_{7/2}\frac{7}{2}\rangle$
29.494	-10.88	$ 503\frac{7}{2}\rangle$	$79.3\% 503\frac{7}{2}\rangle + 7.8\% 523\frac{7}{2}\rangle + 5.5\% 303\frac{7}{2}\rangle$
		$ 5f_{7/2}\frac{7}{2}\rangle$	$85.9\% 5f_{7/2}\frac{7}{2}\rangle + 5.9\% 3f_{7/2}\frac{7}{2}\rangle + 4.3\% 7f_{7/2}\frac{7}{2}\rangle$
	-10.72	$ 505\frac{9}{2}\rangle$	$88.3\% 505\frac{9}{2}\rangle + 11.4\% 514\frac{9}{2}\rangle + 0.2\% 1105\frac{9}{2}\rangle$
		$ 5h_{9/2}\frac{9}{2}\rangle$	$99.5\% 5h_{9/2}\frac{9}{2}\rangle + 0.2\% 11h_{9/2}\frac{9}{2}\rangle + 0.1\% 5h_{11/2}\frac{9}{2}\rangle$
34.0	-11.07	$ 503\frac{7}{2}\rangle$	$78.6\% 503\frac{7}{2}\rangle + 10.0\% 523\frac{7}{2}\rangle + 5.9\% 303\frac{7}{2}\rangle$
		$ 5f_{7/2}\frac{7}{2}\rangle$	$87.3\% 5f_{7/2}\frac{7}{2}\rangle + 6.2\% 3f_{7/2}\frac{7}{2}\rangle + 4.6\% 7f_{7/2}\frac{7}{2}\rangle$
	-10.29	$ 505\frac{9}{2}\rangle$	$88.5\% 505\frac{9}{2}\rangle + 11.1\% 514\frac{9}{2}\rangle + 0.2\% 1105\frac{9}{2}\rangle$
		$ 5h_{9/2}\frac{9}{2}\rangle$	$99.5\% 5h_{9/2}\frac{9}{2}\rangle + 0.2\% 11h_{9/2}\frac{9}{2}\rangle + 0.1\% 9h_{9/2}\frac{9}{2}\rangle$

Prediction: "Stability Inversion" in ^{162}Pt



- Extrapolating the linear trends to $Z=78$ reveals a striking crossing point: stability Inversion occurs in ^{162}Pt : The excited isomeric state outlives its ground state
- The high- K configuration severely hinders it via two mechanisms:
 1. Lower Preformation Probability:
The two-quasiparticle broken-pair state decreases the pairing density, making it harder for an α -cluster to form.
 2. Reduced Penetration Probability:
The 8^+ state confronts a much higher and wider potential barrier

the scope of the present work. However, similar to the superheavy mass region [5], one can still conclude from the facts mentioned above that the high- K isomer may enhance the survival probabilities of drip-line nuclei with sufficiently short lifetimes and help to extend the nuclear chart. It seems that $^{162}_{78}\text{Pt}_{84}$ could be the possible candidate to probe the lifetime inversion between ground and isomeric states in the proton drip-line region.

- Investigated the properties of the proton drip-line nucleus ^{160}Os using configuration-constrained PES calculations in $(\beta_2, \gamma, \beta_4)$ space.
- Demonstrated that the structure of the $K^\pi = 8^+$ isomer is highly sensitive to spin-orbit coupling λ uncertainty. Variations in λ cause orbital inversion ($f_{7/2}$ and $h_{9/2}$) and severe wave-function mixing, making arbitrary configuration assignments extremely risky.
- Calculations successfully reproduced the experimental excitation energy (1.844 MeV) and oblate shape , revealing the critical role of pairing interactions and nucleon polarization.
- Proposed the undiscovered drip-line nucleus ^{162}Pt as a prime candidate to experimentally verify "Stability Inversion" in the future.

Thank you for your attention!